

DistriNVS: Distributional Geometry Transport and Visibility-Constrained Completion for Cross-Endoscope Novel-View Synthesis

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Abstract—Cross-endoscope novel-view synthesis must both transport tissue observed by the source camera and infer tissue revealed only from the target viewpoint. These two regimes are confounded when sparse or erroneous depth is projected as deterministic points: sampling gaps, depth failures, surface collisions, and genuine disocclusions all become visually identical holes. We present DistriNVS, a reliability-aware framework that models every source depth as a distribution and propagates it through calibrated geometry. Its distributional surface splatting renderer produces an anisotropic target-plane footprint together with support, depth variance, and collision entropy. These statistics drive a transport-or-synthesize router: a residual transport expert corrects co-visible appearance, whereas an uncertainty-conditioned Fourier network synthesizes unsupported or geometrically unreliable content. A final hard composition prevents the network from modifying trustworthy projected pixels. Joint target RGB-depth prediction, backward reprojection, and an explicit risk head provide geometric and failure-awareness constraints. The prescribed protocol trains on a fixed set of 13 iMED scenes and evaluates without adaptation on held-out EndoVis datasets 8 and 9 using a strict raw-overlap protocol.

I. INTRODUCTION

Robot-assisted minimally invasive surgery depends on endoscopic vision for scene perception and instrument guidance [1], [2]. Cross-endoscope novel-view synthesis can broaden this visual context by predicting a target-endoscope view from synchronized appearance and geometry acquired by a spatially displaced source endoscope [3]. Unlike left-to-right synthesis within one stereo endoscope, the target may observe anatomy absent from both source cameras. The task therefore combines two epistemically distinct regimes: co-visible anatomy has measured appearance that should be transported through calibrated geometry, whereas target-only anatomy lacks source evidence and must be synthesized. A reliable system should preserve supported measurements, identify inferred content, and thereby support wider-field robotic perception, retrospective visualization, and surgical simulation [4]. Such uncertainty-aware cross-view completion may also support visual-state construction for surgical robot world models [5].

Achieving this separation is difficult because endoscopic depth and calibration are unreliable around specularities, weak texture, depth discontinuities, instruments, and deforming tissue [6]. Conventional forward warping represents each valid source pixel by one 3-D point; discretized projection leaves sampling gaps, while multiple points may compete for the same target pixel. A binary validity mask then conflates

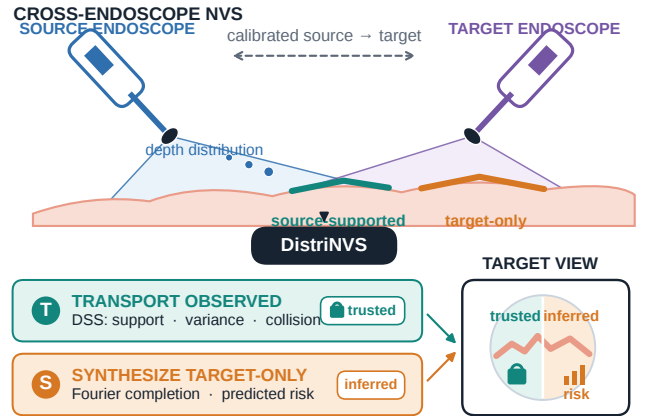


Fig. 1. Cross-endoscope synthesis with explicit provenance. DSS transports source-supported anatomy and exposes visibility uncertainty; DistriNVS synthesizes target-only content while trusted composition preserves reliable projected RGB-D.

missing depth, rasterization holes, depth-ordering conflicts, and genuine cross-endoscope disocclusions. Treating these cases uniformly either degrades observed structure or permits hallucination where geometric evidence remains trustworthy.

Existing methods address complementary parts of this problem but do not expose the required evidence boundary. Differentiable point rendering and softmax splatting improve feature transport and collision handling [7], [8], while reference-guided inpainting combines depth-based alignment with learned completion [9]. Large-mask inpainting networks provide strong global image priors [10], [11], and endoscopic video completion adds depth, temporal, or stereo cues [12]–[14]. Endoscopic neural rendering has likewise advanced sparse-view reconstruction and uncertainty modeling [15], [16]. However, deterministic projection commonly assigns equal reliability to all valid depths, while image-space uncertainty does not explicitly describe how source geometry supports a spatially offset target view. Neither approach propagates per-pixel depth uncertainty through the cross-endoscope transform and uses the resulting visibility statistics to constrain where appearance may be synthesized.

Our central premise is to *transport reliable geometric evidence and synthesize only unsupported or geometrically unreliable regions*. We introduce DistriNVS, a reliability-aware framework for direct cross-endoscope novel-view synthesis. Its distributional surface splatting (DSS) represents each source depth as a distribution whose bounded mean

and scale, surface normal, and confidence are predicted by a depth reliability encoder. DSS projects deterministic sigma samples through the calibrated cross-endoscope transform and aggregates source features using Jacobian- and normal-aware footprints with soft nearest-surface visibility, yielding target support, depth variance, and collision entropy. These statistics drive a continuous gate between residual transport for co-visible regions and uncertainty-conditioned Fourier completion for unsupported or ambiguous regions. A trusted mask hard-copies trusted DSS-supported RGB-D into the output, while target-depth and risk prediction plus backward RGB-D reprojection constrain completion during training.

The paper makes three contributions:

- We formulate cross-endoscope novel-view synthesis as a reliability-aware separation between transport of anatomy observed by the source endoscope and constrained inference of content revealed only by the target endoscope.
- We introduce DSS, a differentiable renderer that propagates depth distributions through calibrated cross-endoscope geometry and exposes target support, depth variance, and collision entropy beyond binary validity.
- We design a two-expert decoder that routes co-visible regions to residual transport and unsupported or geometrically unreliable regions to uncertainty-conditioned Fourier completion, while trusted-pixel composition preserves reliable projected RGB-D.

II. RELATED WORK

We briefly review image completion, geometry-guided view transport, and endoscopic neural rendering, with particular attention to how existing approaches represent missing content, exploit calibrated geometry, and quantify uncertainty.

A. Image and Endoscopic Video Inpainting

Generic image inpainting combines mask-aware feature extraction with increasingly broad appearance priors. Partial and gated convolutions prevent invalid activations from contaminating local features [17], [18], whereas LaMa and UFFC use Fourier operations for global context and MAT restricts attention to valid tokens [10], [11], [19]. Recent diffusion models such as BrushNet and TurboFill incorporate masked-image features into pretrained generators or adapt few-step sampling for efficient completion [20], [21]. Despite strong appearance priors, these methods define holes in a fixed image coordinate system and do not describe geometric provenance. Endoscopic completion adds domain cues: temporal learning removes specularities, DAEVI combines spatio-temporal RGB and depth, and SSIFNet fuses flow and stereo-temporal information for instrument occlusions [12]–[14]. Such methods exploit neighboring frames or views but still restore defects defined in the original camera coordinates. Cross-endoscope synthesis instead creates holes through viewpoint transfer: target-only anatomy is unobserved, while rasterization gaps or depth conflicts may retain source support. DistriNVS therefore uses calibrated

projection statistics to distinguish true disocclusions from rasterization gaps and depth conflicts.

B. Geometry-Guided Transport and Completion

Geometry-guided view synthesis preserves correspondence by transporting source evidence before generating missing content. SynSin predicts and projects a feature point cloud, softmax splatting resolves many-to-one mappings with learned importance, and GeoFill refines depth and pose before reprojection and inpainting [7]–[9]. Recent feed-forward Gaussian renderers use stronger explicit representations: MVSplat localizes Gaussians with plane-sweep cost volumes, whereas DepthSplat combines multi-view correspondence with monocular depth priors [22], [23]. They reduce per-scene optimization but infer a radiance representation from multiple calibrated observations, without an explicit per-pixel boundary between measured and synthesized appearance. Generative methods target wider gaps. LeftRefill frames reference-conditioned synthesis as paired-canvas completion, MultiDiff conditions video diffusion on depth-warped reference views, and point-guided latent diffusion uses stereo-derived clouds for laparoscopic view synthesis [24]–[26]. These models can generate plausible structures in large disocclusions, yet plausibility alone does not reveal which pixels remain geometrically supported. DistriNVS instead transports a calibrated source observation with distributional splatting, derives support, variance, and collision statistics during rasterization, and uses them to bound synthesis while preserving trusted pixels.

C. Endoscopic Novel-View Synthesis and Uncertainty

Endoscopic neural rendering must handle non-rigid tissue, view-dependent illumination, tool occlusions, and sparse camera coverage. EndoNeRF represents deformation with a neural radiance field, while EndoSurf combines dynamic modeling with a neural surface [27], [28]. EndoSparse introduces geometric and appearance priors into Gaussian splatting, and Endo-GSMT uses depth, dense-displacement, and motion priors for dynamic reconstruction [15], [29]. GastroNVS further shows that large viewpoint changes in real gastroscopy amplify geometric artifacts and inter-view illumination mismatch [30]. These systems build persistent scene representations from multiple frames and principally address reconstruction fidelity within a captured scene. Uncertainty-aware approaches address sparse evidence from another direction: UC-NeRF conditions a radiance field on multi-view uncertainty, whereas ExtraGS uses uncertainty to sample virtual cameras and weight diffusion-refined pseudo-observations for extrapolation [4], [16]. Thus, neither uncertainty nor generative priors are new to endoscopic rendering. The distinction is operational: DSS propagates a distribution for each input depth through the calibrated cross-endoscope transform and derives target support, depth variance, and collision entropy from the same rasterization that transports appearance. DistriNVS uses these quantities to route completion and protect trusted measurements during unseen-sequence inference without target-scene optimization.

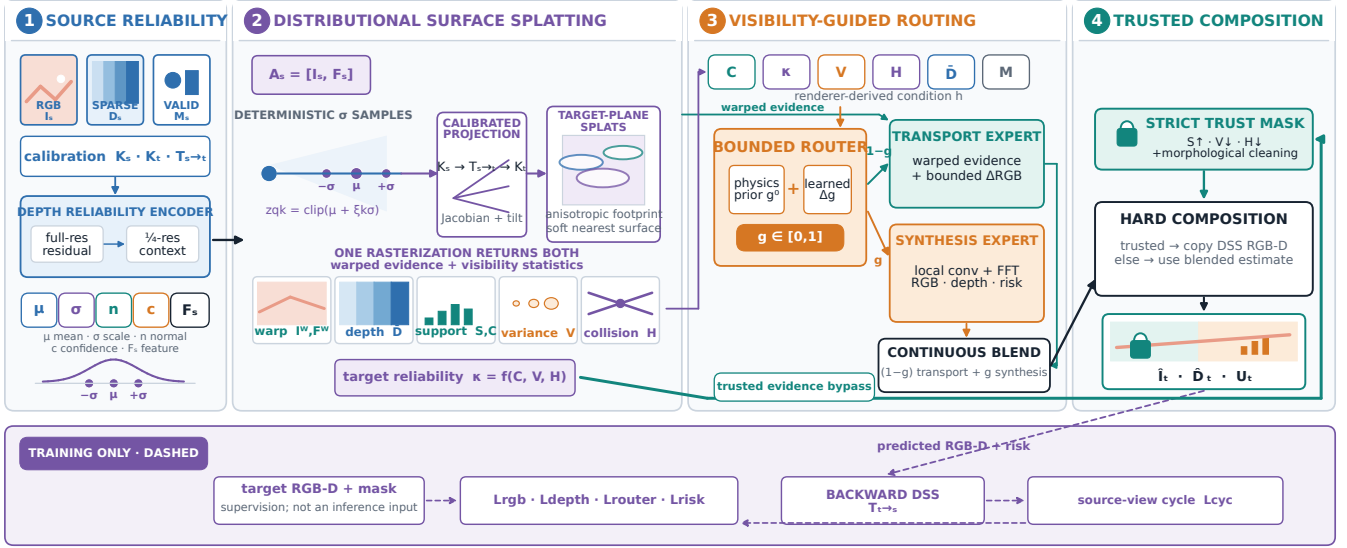


Fig. 2. DistriNVS framework. A reliability encoder predicts source geometry and features. DSS transports deterministic depth hypotheses through the calibrated cross-endoscope transform and jointly produces warped evidence and target-plane visibility statistics. These statistics condition a bounded transport-or-synthesize gate, while an independent trusted mask hard-copies accepted DSS evidence. Solid arrows denote inference; the dashed lower lane is used only during training.

III. METHOD

A. Problem Formulation and Overview

Given a synchronized, calibrated cross-endoscope pair, DistriNVS maps a source RGB image I_s , sparse source depth D_s , its validity mask M_s , camera intrinsics K_s, K_t , and the rigid transform $T_{s \rightarrow t} \in SE(3)$ to target RGB $\hat{I}_t$, target depth $\hat{D}_t$, and a per-pixel risk proxy U_t . The learned mapping is parameterized by θ . The synchronized target RGB-D pair and target instrument mask are used only for training losses or evaluation; none is an inference input.

DistriNVS follows four stages (Fig. 2). First, a depth reliability encoder converts sparse RGB-D into a source feature map and a per-pixel depth distribution. Second, distributional surface splatting (DSS) transports RGB and features while exposing support, depth variance, and collision entropy in the target view. Third, these renderer statistics control a mixture of a transport expert and a synthesis expert. Finally, a spatially consistent trusted mask hard-copies reliable rendered evidence into the output. A depth-guided backward photometric cycle and the risk objective are active only during training.

B. Reliability-Aware Source Geometry

Naive depth completion assigns the same status to measured and inferred geometry. To retain this distinction, the reliability encoder processes $[I_s, \tilde{D}_s, M_s]$, where $\tilde{D}_s$ is the valid sparse depth normalized by its per-sample median s_D ; missing pixels are set to zero. During training, M_s is the post-dropout input mask and M_s^{gt} is the pre-dropout supervision mask. The encoder predicts a source feature F_s , depth mean μ , uncertainty scale σ , confidence c , and a unit orientation proxy n . At source pixel q , measured depth receives a bounded multiplicative correction to $D_s(q)$,

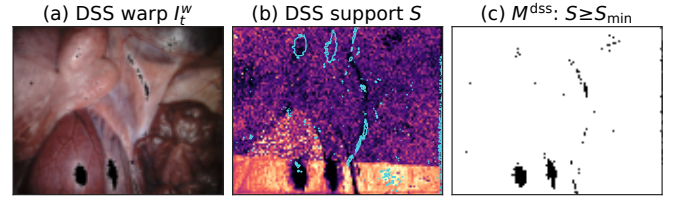


Fig. 3. How continuous DSS support becomes a valid warp, illustrated on a zero-shot EndoVis dataset 8 result. (a) The completed-geometry DSS warp; unsupported pixels are black. (b) Unnormalized support S , with colors capped at the framewise 99th percentile only for display; the cyan contour marks $S_{\min}$. (c) The exact model mask M^{dss} obtained from the uncapped support; white pixels satisfy $S_p \geq S_{\min}$.

whereas missing pixels use a positive completion branch rescaled by s_D . Both branches yield μ_q within the physical depth interval, while a separate bounded output gives σ_q . The confidence parameterization ensures that observed geometry always has higher confidence than completed geometry: completion can add support without acquiring measurement-level authority. The orientation proxy is normalized with a source-camera optical-axis prior and is learned only through downstream losses; it is not presented as a metric surface normal.

C. Distributional Surface Splatting

Given the appearance-feature tensor $A_s = [I_s, F_s]$, DSS renders $[I_t^w, F_t^w]$ and an expected target depth $\hat{D}_t$. Unlike a binary warp, it also returns support S , coverage C , depth variance V , collision entropy $\mathcal{H}$, target confidence κ , and a valid-render mask M^{dss} . Its key distinction from ordinary forward warping is that every source pixel transports a compact depth distribution rather than a single point.

Depth hypotheses. DSS uses three deterministic samples and transports each one through the calibrated source-to-target

camera map:

$$\begin{aligned} z_{qk} &= \text{clip}_{[D_{\min}, D_{\max}]}(\mu_q + \xi_k \sigma_q), \\ (\mathbf{x}_{qk}, z_{qk}^t) &= \Pi_{s \rightarrow t}(q, z_{qk}), \quad \xi_k \in \{-1, 0, 1\}. \end{aligned} \quad (1)$$

Here, a_k is the fixed quadrature weight of sample k , $\mathbf{x}_{qk} = (u_{qk}, v_{qk})$ and z_{qk}^t are its target coordinate and target-camera depth, and $\Pi_{s \rightarrow t}$ denotes back-projection by $\mathbf{K}_s$, transformation by $\mathbf{T}_{s \rightarrow t}$, and projection by $\mathbf{K}_t$ under the standard zero-skew pinhole model [7], [8].

Uncertainty-aware visibility. Depth uncertainty is converted into a sample-dependent, axis-aligned target footprint. Let $J_{qk}^\alpha = |\partial \alpha_{qk} / \partial z|$ be the projected displacement per unit source depth along $\alpha \in \{u, v\}$. Its scale is

$$r_{qk}^\alpha = \text{clip}_{[0.35, \sigma_{\text{px}}^{\max}]}[b + J_{qk}^\alpha \sigma_q + b_n(1 - |n_q^z|)]. \quad (2)$$

Here, b is the base footprint, b_n expands grazing surfaces, n_q^z is the optical-axis component of $\mathbf{n}_q$, and $\sigma_{\text{px}}^{\max}$ is the maximum pixel-space scale. For target pixel $p = (u_p, v_p)$, define the normalized squared offset $\rho_{qkp}^2 = \sum_{\alpha \in \{u, v\}} [(\alpha_{qk} - \alpha_p) / r_{qk}^\alpha]^2$. The contribution of a projected hypothesis is

$$w_{qk \rightarrow p} = c_q a_k \exp \left[-\frac{\rho_{qkp}^2}{2} - \frac{[z_{qk}^t - z_{\min}(p)]_+}{\tau_z} \right]. \quad (3)$$

Here, c_q is source confidence, $z_{\min}(p)$ is the detached nearest arriving depth, $[x]_+ = \max(x, 0)$, and τ_z controls soft occlusion. The minimum and integer assignment are discrete; the remaining quantities are differentiable.

Reliability statistics. Let Ω_p be the set of sample contributions reaching p , with $i = (q_i, k_i)$, source pixel q_i , weight w_i , and depth z_i . Set $S_p = \sum_{i \in \Omega_p} w_i$ and $\tilde{S}_p = \max(S_p, \epsilon_{\text{dss}})$. For any sample attribute f_i , define the DSS aggregation operator

$$\langle f \rangle_p^w = \sum_{i \in \Omega_p} \frac{w_i}{\tilde{S}_p} f_i. \quad (4)$$

Applying Eq. (4) to $\mathbf{A}_s(q_i)$, z_i , and $[z_i - \bar{D}_t(p)]^2$ gives $\mathbf{A}_t^w(p)$, $\bar{D}_t(p)$, and V_p , respectively, where $\mathbf{A}_t^w = [\mathbf{I}_t^w, \mathbf{F}_t^w]$ and ϵ_{dss} prevents division by zero. Support is mapped to coverage $C_p = 1 - \exp(-S_p)$. Let $\pi_i = w_i / \tilde{S}_p$, $N_p = |\Omega_p|$, and $H_N(\pi_p) = -\sum_{i \in \Omega_p} \pi_i \log \pi_i / \log \max(N_p, 2)$ denote normalized weight entropy. Because this entropy alone rises under harmless sub-pixel overlap, DSS modulates it by depth dispersion:

$$\mathcal{H}_p = H_N(\pi_p) [1 - \exp(-V_p / \tau_z^2)]. \quad (5)$$

Target confidence κ_p is the product of coverage C_p , variance attenuation $\exp[-V_p / (4\tau_z^2)]$, and entropy attenuation $\exp(-\mathcal{H}_p)$. Thresholding support at $S_{\min}$ yields the valid-render mask M^{dss} visualized in Fig. 3; a pixel is valid exactly when $S_p \geq S_{\min}$. Where this mask is zero, DSS sets $\mathbf{A}_t^w$, $\bar{D}_t$, V , $\mathcal{H}$, and κ to zero while retaining the raw support S and coverage C . Thus, V_p and $\mathcal{H}_p$ preserve evidence of uncertain depth ordering instead of reducing reliability to a binary hole mask.

D. Reliability-Guided Completion and Preservation

Both experts receive $[\mathbf{I}_t^w, \mathbf{F}_t^w, \mathbf{h}]$, where the six-channel condition $\mathbf{h}$ contains coverage, confidence, normalized variance, collision entropy, normalized target depth, and DSS validity, in that order. A physics prior g_p^0 favours synthesis when confidence falls or geometric ambiguity rises, and is set to one outside DSS support. We clip this prior by a margin ϵ_g and denote its stop-gradient logit by ℓ_p^0 . The learned router can then make only a bounded correction:

$$g_p = \text{sigmoid}[\ell_p^0 + \eta_g \tanh R_\theta(\mathbf{h})_p]. \quad (6)$$

Here, ϵ_g is the clipping margin, η_g bounds the learned correction, and R_θ is the residual router. The gate convexly blends the transport and synthesis RGB-depth predictions: $g_p = 0$ selects transport and $g_p = 1$ selects synthesis. We denote the two blended outputs by $(\mathbf{I}_t^{\text{mix}}, \mathbf{D}_t^{\text{mix}})$ and use $\odot$ for element-wise multiplication below.

The transport expert produces $\mathbf{I}_t^{\text{trans}}$ by adding a ± 0.15 bounded RGB residual to $\mathbf{I}_t^w$ and clipping the result to $[0, 1]$. The synthesis expert is a four-resolution U-Net that fuses local 3×3 convolution and real-FFT branches, conditioned at every resolution by $\mathbf{h}$. Its five output channels use sigmoid RGB; the depth channel passes through softplus, receives a 0.05 positive offset, and is multiplied by s_D ; the risk channel uses a sigmoid. These channels produce $(\mathbf{I}_t^{\text{syn}}, \mathbf{D}_t^{\text{syn}}, \mathbf{U}_t^{\text{syn}})$. This local-global design follows Fourier inpainting networks [10], [11], but its weights are trained within DistriNVS rather than imported from either model. The risk output is treated as an error-scale proxy, not a calibrated failure probability.

Soft routing alone cannot guarantee preservation. A pixel enters the initial mask $M^{\text{trust}, 0}$ only when it has valid DSS support, $S_p \geq S_{\text{trust}}$, $V_p \leq V_{\text{trust}}$, and $\mathcal{H}_p \leq \mathcal{H}_{\text{trust}}$. We apply a 7×7 binary opening to its untrusted complement and intersect the complement of that result with M^{dss} , yielding the spatially cleaned mask M^{trust} . The router zeros g on $M^{\text{trust}, 0}$; hard composition then makes the effective gate zero over the cleaned M^{trust} . Let $\hat{\mathbf{Z}}_t = [\hat{\mathbf{I}}_t, \hat{\mathbf{D}}_t, \mathbf{U}_t]$, $\mathbf{Z}_t^w = [\mathbf{I}_t^w, \bar{D}_t, 0]$, and $\mathbf{Z}_t^{\text{mix}} = [\mathbf{I}_t^{\text{mix}}, \mathbf{D}_t^{\text{mix}}, \mathbf{U}_t^{\text{syn}}]$, with binary masks broadcast over channels. The three final outputs share the single preservation rule

$$\hat{\mathbf{Z}}_t = M^{\text{trust}} \odot \mathbf{Z}_t^w + (1 - M^{\text{trust}}) \odot \mathbf{Z}_t^{\text{mix}}. \quad (7)$$

Thus, trusted output is exactly equal to the accepted DSS warp—not necessarily ground truth. Zero drift relative to that warp is an architectural constraint.

Reported inference additionally repairs 8-connected untrusted components that each occupy at most 2% of the image. Radius-3 Telea fast-marching inpainting [31] is applied to $\mathbf{I}_t^w$ only on the recorded repair mask; its values replace the corresponding untrusted output pixels. Larger components remain learned predictions, trusted pixels are never changed, and the repair is absent during training.

E. Learning Objectives

Let M_{tis} exclude target instrument pixels when a tool mask is available. The untrusted and trusted training regions are

$M_u = (1 - M^{\text{trust}}) \wedge M_{\text{tis}}$ and $M_v = M^{\text{trust}} \wedge M_{\text{tis}}$, respectively. The seam mask M_b is the morphological boundary of $1 - M^{\text{trust}}$ generated by a 7×7 dilation-erosion window and intersected with M_{tis} . Charbonnier reconstruction terms $\mathcal{L}_h$, $\mathcal{L}_v$, and $\mathcal{L}_b$ act on M_u , M_v , and M_b , respectively, while $\mathcal{L}_{\nabla}$ matches image gradients on $M_u \cup M_b$. This model-defined training region is distinct from the raw-warp hole mask used for evaluation.

Available source depth supervises (μ, σ) on the pre-dropout mask $\mathbf{M}_s^{\text{gt}}$ through a masked Laplace negative log-likelihood $\mathcal{L}_{d\text{-nll}}$; optional target depth uses a masked relative- L_1 term $\mathcal{L}_{t\text{-depth}}$. For the depth-guided cycle, $(\hat{\mathbf{I}}_t, \hat{\mathbf{D}}_t)$ is rendered back with $\mathbf{T}_{t \rightarrow s} = \mathbf{T}_{s \rightarrow t}^{-1}$. Detached risk increases the backward depth scale up to its configured maximum, and confidence is $1 - \mathbf{U}_t$ with a floor of 0.05. The resulting $\mathcal{L}_{\text{cyc}}$ is a masked source-view Charbonnier RGB loss; no separate depth-cycle loss is used.

The router target is one whenever support, variance, or collision entropy fails its trust threshold and zero otherwise; $\mathcal{L}_{\text{router}}$ is binary cross-entropy on the pre-sigmoid gate logit. To give risk a statistical meaning, define the mean RGB absolute error $e_p = \|\hat{\mathbf{I}}_t(p) - \mathbf{I}_t(p)\|_1/3$ and the positive Laplace scale $s_p = 0.01 + 0.49\mathbf{U}_t(p)$. The risk objective is

$$\mathcal{L}_u = \langle e_p/s_p + \log s_p \rangle_{M_u}. \quad (8)$$

Here, $\mathbf{I}_t$ is the training target and $\langle \cdot \rangle_M$ denotes the mean over pixels selected by M and, for channel-valued arguments, over their channels. For a compact presentation, let $\mathcal{L}_{\text{app}} = \lambda_h \mathcal{L}_h + \lambda_v \mathcal{L}_v + \lambda_b \mathcal{L}_b + \lambda_{\nabla} \mathcal{L}_{\nabla}$ and $\mathcal{L}_{\text{depth}} = \mathcal{L}_{d\text{-nll}} + \mathcal{L}_{t\text{-depth}}$. The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{app}} + \lambda_d \mathcal{L}_{\text{depth}} + \lambda_c \mathcal{L}_{\text{cyc}} + \lambda_r \mathcal{L}_{\text{router}} + \lambda_u \mathcal{L}_u. \quad (9)$$

All λ terms are non-negative loss weights; their implemented values are given in the experimental setup.

The learned inference path ends at Eq. (7), followed only by the disclosed deterministic repair in reported runs; target supervision, tool masks, and the backward cycle are not required. The current formulation assumes known zero-skew pinhole intrinsics and a rigid inter-endoscope transform, and it uses an axis-aligned approximation to the projected uncertainty footprint. These choices define the scope of the method and are tested separately from the learned completion model in the ablations.

IV. EXPERIMENTS

A. Datasets and Zero-Shot Protocol

Training uses the frozen list of 13 iMED Task 2 scene directories supplied in the project configuration [32]. Splits are made at scene level, not at frame level. The training loader uses a synchronized source RGB-depth and target RGB pair at 640×512 resolution. Fifteen percent of valid source depth samples are removed online, with additional 9×9 missing islands, so that the reliability encoder observes both isolated and contiguous depth failures. EndoVis/SCARED datasets 8 and 9 are reserved for final zero-shot evaluation; neither their images nor metrics are used for model selection.

We determine inclusion using a renderer-only pass from the raw valid source depth. A frame is evaluated only if $|M_{\text{raw}}^{\text{dss}}|/(H_I W_I) > 0.5$; the learned depth completion, target image, and target tool mask cannot influence this decision. This reproduces the project’s requested strict overlap criterion and prevents a learned module from selecting favourable frames. The current tables report frame-wise means and sample standard deviations, both rounded to two decimal places. Paired sequence-level confidence intervals and permutation tests will be added once all baseline predictions are frozen.

[TODO: Verify the final dataset provenance, license statement, total training pair count, held-out sequence count, and the immutable list of selected test frames before submission.]

B. Baselines, Metrics, and Implementation

Geometry baselines will include hard point projection, nearest-neighbour small-hole filling, deterministic soft splatting, and softmax splatting. Image completion baselines will include Telea, LaMa, UFFC, MAT, Stable Diffusion 1.5 Inpainting, and the existing LCM+endoscopy-LoRA system. Directly relevant endoscopic baselines (DAEVI and SSIFNet) will be evaluated when code and an equivalent cross-view protocol are available; otherwise their non-comparable reported values will not be inserted into the main table. Every completion method will use the same raw mask, spatial resolution, selected frames, and known-pixel hard composition.

The existing evaluator reports full-, raw-hole-, raw-visible-, and 7-pixel seam-band PSNR/SSIM, full-image and hole-region LPIPS, known-region drift, risk expected calibration error (ECE), area under the risk-coverage curve (AURC), and a projection-defect taxonomy. The final study will additionally report DISTs, depth AbsRel/RMSE, disparity endpoint error, normal error, latency, throughput, peak memory, and parameter count. Results will be stratified by raw overlap, hole ratio, depth validity, instrument presence, and sequence.

The reference configuration uses sigma offsets $(-1, 0, 1)$ with quadrature weights $(0.2741, 0.4518, 0.2741)$, a one-pixel splat radius, $\tau_z = 2$ mm, $S_{\text{min}} = 0.03$, $S_{\text{trust}} = 0.35$, $V_{\text{trust}} = 9$ mm², $\mathcal{H}_{\text{trust}} = 1.0$, a 7-pixel seam band, and base width 32. The reliability encoder combines a full-resolution residual branch with a quarter-resolution context branch; when no valid source depth is available, it uses $s_D = 100$ mm. Valid source depth is normalized as $\tilde{D}_s = \text{clip}(D_s/s_D, 0, 5)$, and missing pixels are set to zero. Measured and completed confidence lie in $[0.5, 1]$ and $[0, 0.35]$, respectively. The measured-depth residual ratio is bounded by 0.1, and the positive completion branch uses an offset of 0.05. Depth and uncertainty are bounded to $[1, 500]$ mm and $[0.25, 20]$ mm, respectively; $b = b_n = 0.75$ px, $\sigma_{\text{px}}^{\text{max}} = 2.5$ px, and the Charbonnier constant is 10^{-3} . During sparse accumulation, hypotheses contribute only to the radius-one neighbourhood of their integer target coordinate; spatial and final weights no greater than 10^{-5} and 10^{-8} are pruned, and $\epsilon_{\text{dss}} = 10^{-8}$ stabilizes normalization. The

router uses $\bar{V} = V/(V + \tau_z^2)$ and $\tilde{D} = \text{clip}(\bar{D}_t/s_D, 0, 5)/5$ as its normalized variance and depth channels. Its prior is $g^0 = 0.65(1 - \kappa) + 0.20\bar{V} + 0.15\mathcal{H}$, with $\epsilon_g = 10^{-4}$ and $\eta_g = 0.25$. Because $\mathcal{H} \in [0, 1]$, this reference threshold is neutral in the hard trusted mask; collision entropy still affects κ and the soft synthesis gate. Reported inference uses the disclosed 2%-area, radius-3 Telea repair for small untrusted components. Training is scheduled for 10,000 optimizer steps with AdamW, learning rate 2×10^{-4} , weight decay 10^{-4} , and three-GPU distributed training. The per-GPU batch size is 1 with four-step gradient accumulation, giving an effective batch size of 12. We use mixed precision, a maximum gradient norm of 1, and seed 6666. The implemented loss weights ($\lambda_h, \lambda_v, \lambda_b, \lambda_\nabla, \lambda_d, \lambda_c, \lambda_r, \lambda_u$) are (3, 1, 2, 0.2, 0.2, 0.2, 0.1, 0.1).

[TODO: Record the final checkpoint, model-selection rule, GPU model, training time, inference warm-up, precision, exact parameter/FLOP counts, and all baseline versions and checkpoints.]

C. Main Comparison

We first compared all methods on the same 1,392-frame iMED validation set. To avoid method-dependent masks, every prediction was re-evaluated against the same target image and raw-warp hole mask. The mask covered 25.84% of an image on average. We additionally re-evaluated four scene-optimized Gaussian renderers from the comparison benchmark; each retained method produced a prediction for every selected frame. These Gaussian methods use the dataset-provided ground-truth depth and are therefore reference reconstruction baselines rather than matched feed-forward completion models. PR-ENDO, EndoGS, and EndoGaussian were omitted because their available predictions cover only 1/7, 5/7, and 6/7 validation scenes, respectively. Table I reports the frame-wise mean and sample standard deviation under this common protocol. The deterministic small-component repair affected $1.17 \pm 0.94\%$ of image pixels per validation frame on average; the 2% criterion applies to each connected component, so their union may be larger.

DistriNVS achieved the best distortion- and structure-based result, improving full-image PSNR by 0.94 dB and SSIM by 0.05 over the strongest corresponding image-completion baselines. When the Gaussian reconstruction baselines are included, the SSIM margin narrows to 0.02 over Free-SurGS, whereas the PSNR margin remains 0.94 dB over the LCM baseline. The advantage was larger inside the missing region: hole PSNR increased by 3.57 dB over LaMa, while hole SSIM increased by 0.08 over the supervised LoRA baseline. However, LaMa obtained lower full-image LPIPS (0.47 ± 0.03 versus 0.55 ± 0.03) and lower hole LPIPS (0.48 ± 0.04 versus 0.52 ± 0.03). The results therefore show stronger pixel and structural fidelity for DistriNVS, but not a uniform advantage in learned perceptual distance.

We separately evaluated DistriNVS on held-out EndoVis Datasets 8 and 9 using the strict raw overlap > 0.5 inclusion

rule. Among 124 candidate frames, 78 met the rule. Table II reports the two datasets separately and jointly.

Across the 78 accepted frames, DistriNVS obtained 23.79 ± 2.49 dB PSNR, 0.68 ± 0.12 SSIM, 0.23 ± 0.06 LPIPS, 21.82 ± 3.81 dB hole PSNR, 0.61 ± 0.15 hole SSIM, and 0.28 ± 0.09 hole LPIPS. The deterministic repair affected $3.75 \pm 2.11\%$ of pixels per accepted frame. Known-region drift was zero under the model-defined trusted mask. Baseline inference on Datasets 8 and 9 is not yet complete; consequently, Table II does not support a cross-method zero-shot ranking. Paired confidence intervals and significance tests will be added after the final prediction set is frozen.

D. Ablations and Geometric Diagnostics

The core ablation in Table III follows the implementation flags. The hard-point variant disables depth completion, distributional samples, normal-dependent footprint, physics routing, and the backward cycle. The soft splat variant retains a deterministic Gaussian footprint. Subsequent rows add DSS, the physics router, hard composition, and the RGB-D-defined backward cycle. This ordering tests whether an improvement originates from ordinary anti-aliased splatting or from the proposed distributional visibility model.

[TODO: Add columns for seam error, visible-region PSNR, known drift, depth error, AURC, and runtime. Run focused DSS ablations for 1/3/5 sigma points, isotropic versus anisotropic footprints, normal removal, hard versus soft visibility, and individual $S/V/\mathcal{H}$ router inputs.]

[TODO: Report the paired with/without small-component Telea repair ablation; keep the repair disabled in all architecture-only ablations.]

To determine whether DSS solves the intended geometric problem before learned completion is considered, we will report raw coverage, visible-region RGB error, target-depth error, and occlusion-boundary ordering for hard, softmax, and distributional splatting. The diagnostic label map partitions target pixels into reliable/unclassified support, local sampling gaps, depth-completion- recoverable gaps, unsupported/disoccluded regions, and surface-collision risk. These labels are diagnostics rather than ground truth because a single source view cannot perfectly separate genuine disocclusion from catastrophic depth failure.

[TODO: Insert the DSS-only table and plots showing monotonic relationships between support/variance/collision entropy and actual reprojection error.]

E. Qualitative, Failure, and Efficiency Analysis

Efficiency is central to the ICRA framing. We will measure end-to-end latency including depth reliability estimation, DSS, both experts, and hard composition; DSS-only timing will also be reported to isolate the renderer cost. Failure analysis will cover specular depth loss, calibration perturbation, thin instruments, large disocclusion, weak texture, and dynamic tissue deformation.

TABLE I

UNIFIED COMPARISON ON THE 1,392-FRAME IMED VALIDATION SET. ALL METHODS USE IDENTICAL TARGETS AND RAW-WARP HOLE MASKS. VALUES ARE FRAME-WISE MEAN $\pm$ SD; BEST MEANS ARE BOLD. GAUSSIAN BASELINES ARE SCENE-OPTIMIZED USING DATASET GROUND-TRUTH DEPTH AND HAVE COMPLETE 7/7-SCENE COVERAGE. THE DISTRiNVS ROW INCLUDES THE DISCLOSED SMALL-COMPONENT TELEA REPAIR.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	H-PSNR $\uparrow$	H-SSIM $\uparrow$	H-LPIPS $\downarrow$
<i>Scene-optimized Gaussian reconstruction</i>						
Deform3DGS [33]	15.55 $\pm$ 1.78	0.55 $\pm$ 0.04	0.63 $\pm$ 0.06	14.11 $\pm$ 1.37	0.47 $\pm$ 0.10	0.69 $\pm$ 0.06
Endo-4DGS [34]	16.44 $\pm$ 1.71	0.58 $\pm$ 0.04	0.61 $\pm$ 0.06	15.61 $\pm$ 1.47	0.57 $\pm$ 0.13	0.66 $\pm$ 0.06
Free-SurGS [35]	16.83 $\pm$ 2.11	0.67 $\pm$ 0.06	0.63 $\pm$ 0.04	17.05 $\pm$ 2.38	0.72 $\pm$ 0.04	0.61 $\pm$ 0.04
SurgicalGS [36]	17.41 $\pm$ 1.99	0.60 $\pm$ 0.06	0.60 $\pm$ 0.05	16.98 $\pm$ 2.55	0.61 $\pm$ 0.14	0.65 $\pm$ 0.04
<i>Image completion</i>						
SD1.5 Inpainting	16.84 $\pm$ 2.73	0.57 $\pm$ 0.05	0.52 $\pm$ 0.06	14.56 $\pm$ 3.14	0.50 $\pm$ 0.10	0.64 $\pm$ 0.07
LaMa	18.99 $\pm$ 1.58	0.64 $\pm$ 0.07	0.47 $\pm$ 0.03	20.41 $\pm$ 2.06	0.74 $\pm$ 0.05	0.48 $\pm$ 0.04
LCM + endoscopy LoRA	19.02 $\pm$ 1.64	0.63 $\pm$ 0.09	0.49 $\pm$ 0.03	19.54 $\pm$ 2.28	0.74 $\pm$ 0.09	0.49 $\pm$ 0.03
DistriNVS (ours)	19.96 $\pm$ 1.79	0.69 $\pm$ 0.08	0.55 $\pm$ 0.03	23.98 $\pm$ 3.28	0.82 $\pm$ 0.06	0.52 $\pm$ 0.03

TABLE II

DISTRiNVS ZERO-SHOT PERFORMANCE ON HELD-OUT ENDOVIS DATASETS 8 AND 9. VALUES ARE FRAME-WISE MEAN $\pm$ SD OVER FRAMES SATISFYING raw overlap $>$ 0.5 AND INCLUDE THE DISCLOSED SMALL-COMPONENT REPAIR.

Dataset	Frames	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	H-PSNR $\uparrow$	H-SSIM $\uparrow$	H-LPIPS $\downarrow$
EndoVis 8	48	24.85 $\pm$ 2.00	0.72 $\pm$ 0.12	0.21 $\pm$ 0.04	23.71 $\pm$ 2.74	0.70 $\pm$ 0.09	0.23 $\pm$ 0.06
EndoVis 9	30	22.10 $\pm$ 2.29	0.63 $\pm$ 0.09	0.27 $\pm$ 0.06	18.80 $\pm$ 3.32	0.45 $\pm$ 0.08	0.35 $\pm$ 0.08
Combined	78	23.79 $\pm$ 2.49	0.68 $\pm$ 0.12	0.23 $\pm$ 0.06	21.82 $\pm$ 3.81	0.61 $\pm$ 0.15	0.28 $\pm$ 0.09

[TODO: Replace this placeholder with synchronized columns: source RGB, raw hard warp, raw validity, LaMa, LCM, DistriNVS, target RGB, support, variance, collision entropy, and risk. Include at least one success and one failure for each overlap bin.]

Fig. 4. Planned qualitative comparison and failure visualization. Crops and color scales must be identical across methods; generated and source-backed regions will be explicitly marked.

TABLE III

COMPONENT ABLATION. K IS THE NUMBER OF DEPTH SIGMA POINTS.

Variant	K	DSS	Router	Hard	PSNR
Hard point	1	–	–	✓	–
Soft splat	1	–	–	✓	–
DSS only	3	✓	–	✓	–
DSS + router	3	✓	✓	✓	–
No hard composition	3	✓	✓	–	–
No backward cycle	3	✓	✓	✓	–
Full DistriNVS	3	✓	✓	✓	–

[TODO: Insert the speed/memory/parameter table, calibration-perturbation curves, and blinded expert assessment if available. Do not claim real-time performance until measured on the stated hardware and resolution.]

V. DISCUSSION

Under a common validation protocol, DistriNVS outperformed the evaluated Gaussian reconstruction, generic inpainting, and supervised LoRA baselines in both full-

image and hole-region distortion and structural fidelity. The Gaussian rows provide a reference rather than a matched training comparison, because they are optimized per scene with dataset ground-truth depth. The larger improvement in hole PSNR (3.57 dB) than in full-image PSNR (0.94 dB) is consistent with the intended separation between evidence transport and unsupported-region synthesis. Together with zero measured drift under the trusted mask, these observations support the operational value of constraining where synthesis is permitted. They do not yet isolate the individual contributions of distributional splatting, routing, and hard composition; the planned component ablations remain necessary for that causal claim. Moreover, DistriNVS’s higher LPIPS than LaMa shows that its gains in pixel and structural agreement do not translate uniformly to the AlexNet perceptual feature space; qualitative analysis is needed to determine whether this difference reflects texture realism, colour statistics, or metric bias.

The method does not remove the fundamental ambiguity of unseen tissue. A low risk score indicates model con-

confidence under the training distribution; it does not certify anatomical truth. Moreover, the current formulation assumes known intrinsics and a rigid inter-camera transform for each pair, processes frames independently, and uses an axis-aligned rather than full-covariance footprint. The zero-shot evaluation also revealed a gap between the two held-out datasets: full-image PSNR decreased from 24.85 ± 2.00 dB on EndoVis 8 to 22.10 ± 2.29 dB on EndoVis 9, while hole SSIM decreased from 0.70 ± 0.09 to 0.45 ± 0.08 . This difference limits claims of uniform cross-dataset generalization, and incomplete baseline runs currently prevent a fair cross-method ranking on Datasets 8 and 9. Large calibration errors, non-rigid motion between nominally synchronized frames, or out-of-distribution anatomy may therefore invalidate the transport statistics. Generated output should remain visibly marked and should not be used as a substitute for a live physical endoscopic view.

[TODO: Rewrite the first discussion paragraph from the final ablation and risk curves. Add observed failure modes, not generic limitations, and compare them directly with LaMa/UFCF, DAEVI/SSIFNet, UC-NeRF, and ExtraGS.]

VI. CONCLUSION

This paper introduces a reliability-aware formulation of cross-endoscope novel-view synthesis and implements it through distributional surface splatting, visibility-constrained transport/synthesis routing, trusted-pixel hard composition, and joint appearance–geometry risk modeling. The design turns projection support, depth variance, and collision entropy into operational constraints on what the network may synthesize. **[TODO: After the frozen evaluation, replace the following validation evidence with the frozen test-set comparison if available.]** On a common 1,392-frame validation protocol, DistriNVS exceeded the strongest per-metric baselines by 0.94 dB in full-image PSNR and 3.57 dB in hole PSNR, while attaining a hole SSIM of 0.82 ± 0.06 . These results support visibility-constrained completion under the evaluated setting, although LaMa retained the lowest LPIPS. The difference between EndoVis Datasets 8 and 9 and unfinished zero-shot baseline evaluation define the current boundary of the evidence.

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